



LZAS glass-ceramics for matching 4J50/Cu leads: Crystallization properties and sealing mechanism



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Abstract

This study explores the synthesis of LZAS (Li₂O-ZnO-Al₂O₃-SiO₂) glass-ceramics through fusion and sintering techniques, focusing on the impact of varying the Al₂O₃/ZnO ratio on their crystalline behavior and glass-to-metal sealing properties. Differential scanning calorimetry (DSC), X-ray diffraction (XRD), and scanning electron microscopy (SEM) analyses reveal that increasing the Al₂O₃/ZnO ratio suppresses crystallization while promoting the formation of β -quartz solid solutions and β -spodumene crystalline phases. These changes correlate with a reduction in the coefficients of thermal expansion (CTE) from 13.6×10^{-6} to 10.2×10^{-6} . Additionally, the Al₂O₃/ZnO ratio significantly influences wettability, Vickers hardness, and bending strength. The A4# LZAS glass-ceramic demonstrates compatibility with 4J50/Cu leads, forming a robust chemical bonding layer.

Nevertheless, optimization of the high-temperature sealing conditions remains necessary to achieve improved sealing performance.



Keywords

Glass-ceramic; CTE; 4J50/Cu leads; Matching sealing

1. Introduction

Glass is widely used as a sealing material for various substrates, including glass, ceramics, metals, and semiconductors. Glass-to-metal hermetic sealing technology enables the formation of vacuum-tight electrical connectors, packages, feedthroughs, and electronic components [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#). This technology is renowned for its reliability, meeting the stringent demands of harsh environments and high-performance applications. Recent advancements, particularly in microcrystalline glass sealing techniques, have significantly improved the safety and durability of electronic components [\[5\]](#). With the rapid growth of the electronics industry, metal tube shells now require superior mechanical properties and greater environmental stability [\[6,7\]](#).

Certain electronic devices necessitate alloy leads with high electrical and thermal conductivity [\[8\]](#). However, no single metal or alloy can simultaneously achieve both optimal electrical/thermal performance and effective sealing. For example, while the 4J50 alloy exhibits excellent glass sealing properties, its electrical and thermal conductivity are limited. Copper, on the other hand, offers high electrical and thermal conductivity but struggles with sealing due to its large thermal expansion coefficient and poor compatibility with glass and ceramics [\[9\]](#).

To address this, 4J50 alloy-clad copper rods are utilized as sealing leads, offering a balance between excellent sealing performance and high conductivity [\[10\]](#). However, ensuring compatibility between the thermal expansion coefficients (CTEs) of the glass and metal remains a challenge. Mismatched CTEs can lead to internal stresses during cooling, potentially causing copper core outflow or leakage. LZAS (Li₂O-ZnO-Al₂O₃-SiO₂) glass-ceramics offer a promising solution, with adjustable CTEs, excellent

mobility, mechanical strength, and physicochemical properties [[11], [12], [13], [14]]. These properties make LZAS glass-ceramics ideal for matching 4J50/Cu seals, maintaining structural integrity through thousands of thermal cycles critical for semiconductor and optoelectronic applications [15].

Extensive research has been conducted on LZAS glass-ceramics. For instance, the addition of B_2O_3 has been shown to reduce sintering temperatures, improve thermal stability, lower CTEs, and adjust dielectric properties [16]. Heat treatment conditions and nucleating agents such as P_2O_5 have been found to influence crystallization behavior and the resultant CTEs [17]. Substituting ZnO with Al_2O_3 can significantly lower CTEs [12], while MgO substitution impacts crystallinity, Vickers hardness, flexural strength, and thermal properties [18].

Sealing glass or glass-ceramics typically involves powder or paste forms, with sintering performance playing a critical role in sealing quality. Most studies have focused on the melting method, which does not directly guide the preparation of sealed glass insulators. Gong Wei and collaborators explored the sintering characteristics of LZAS glass-ceramics for surface coatings on Q235 steel substrates [[19], [20], [21]]. However, there is still a lack of systematic investigation into the compositional effects on LZAS sealing glass crystallization and properties. Furthermore, many studies report LZAS glasses with CTEs that are mismatched with 4J50/Cu leads.

This work addresses these gaps by preparing LZAS glass-ceramics using the sintering method and investigating the effects of the Al_2O_3/ZnO ratio on their crystallization behavior and sealing properties.

2. Materials and methods

The experiment utilizes a base system composed of Li_2O , ZnO , Al_2O_3 , SiO_2 , B_2O_3 , and Na_2O , with the nucleating agent P_2O_5 incorporated to investigate the impact of the Al_2O_3/ZnO ratio on the crystallization behavior of the glass-ceramics. The chemical compositions of the A# series glass samples are detailed in Table 1.

Table 1. Chemical composition of A# series glass samples.

Sample	wt%						
	SiO ₂	B ₂ O ₃	Al ₂ O ₃	ZnO	Li ₂ O	Na ₂ O	P ₂ O ₅
A1	50.3	4	4	24	11.1	5	1.6
A2	50.3	4	6	22	11.1	5	1.6
A3	50.3	4	8	20	11.1	5	1.6
A4	50.3	4	10	18	11.1	5	1.6
A5	50.3	4	12	16	11.1	5	1.6

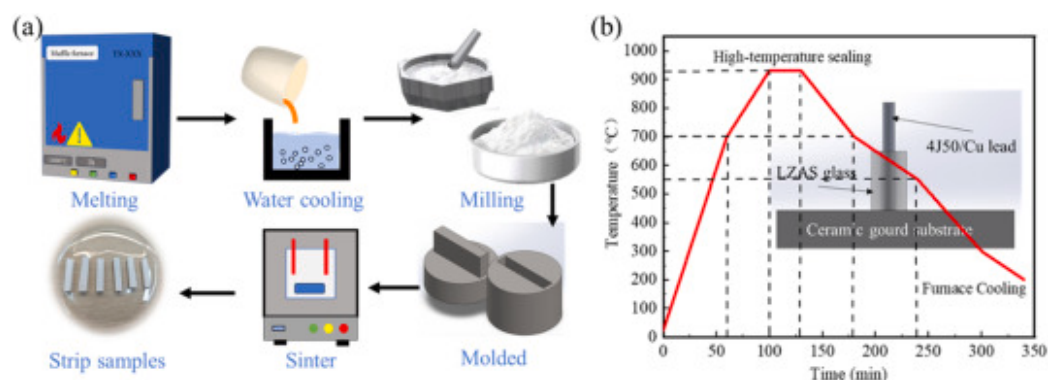
Table 2. Characteristic temperatures and glass-forming abilities determined from DSC curves of A# with a heating rate of 10K/min.

Number	T _g (±1 °C)	T _x (±1 °C)	T _p (±1 °C)	ΔT (±1 °C)	S (±1 °C)
A1	464	536	587	123	7.9
A2	472	541	606	134	9.5
A3	475	549	617	142	10.6
A4	478	558	624	146	11.0
A5	482	566	633	151	11.7

The raw materials used in the experiment were analytical grade (AR) (see [Table 2](#)). Al₂O₃ and ZnO were introduced directly as oxides, SiO₂ was sourced from high-purity quartz sand, B₂O₃ from H₃BO₃, Li₂O from Li₂CO₃, Na₂O from Na₂CO₃, and P₂O₅ from NH₄H₂PO₄. After conversion, the ingredients were weighed using an analytical balance with a precision of 0.01 g. The materials were mixed thoroughly using a mixer for uniform blending, sieved, and sealed for storage.

The glass was melted in an alumina crucible using a high-temperature lift furnace. The process, depicted in [Fig. 1\(a\)](#), involved heating the charge to approximately 1300 °C, then raising the temperature to 1400 °C and maintaining it for 2 h. During the

melting stage, the lift furnace was periodically lowered to ensure uniform mixing through stirring, while the material was inspected to confirm complete melting.



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Fig. 1. (a) Preparation process of LZAS basic glass and glass-ceramic. (b) LZAS glass-ceramic -4J50/Cu Lead Sealing Process.

The molten glass was rapidly quenched in room-temperature water to form glass slag, which was subsequently cleaned, dried, and ball-milled for 70–80 min using an onyx ball mill. The resulting glass powder was sieved and prepared for the sintering method to study its precipitation properties and fabricate the glass-ceramic.

The glass powder was finely ground, weighed (10–15 mg), and placed in an alumina crucible. It was then heated from room temperature to 1000°C at a rate of 10°C/min under a nitrogen atmosphere using a differential scanning calorimeter (DSC, STA449C, Netzsch, Germany) to obtain the DSC curve. This analysis aimed to determine the transition temperature (T_g), onset crystallization temperature (T_x), crystallization peak temperature (T_p), and glass-formation ability (GPA) of the LZAS glass.

The ground glass powder was pressed into flakes and processed using a one-step sintering method at 620°C and 720°C for 2 h to create LZAS glass-ceramic. The resulting material was ground into powder for further characterization. X-ray diffraction (XRD, Bruker D8 Discover, Bruker, Germany) was performed to identify crystalline phases, using a Cu K α target ($\lambda=0.15406$ nm), a scanning range of 10°–70°, a scan rate of 2°/min, and a step size of 0.02°. Jade9 software and standard PDF cards were utilized for phase identification.

The samples were prepared by slicing into thin sections, polishing with 1000-mesh grit, immersing in a 4% HF solution, and sonicating for 1 min. These were subsequently analyzed using field emission scanning electron microscopy (SEM Japan Electronics JSM-IT200) to observe the microscopic morphology of the LZAS glass-ceramic crystals.

The coefficient of thermal expansion (CTE) of the samples was measured using a dilatometer (DIL402C, Netzsch, Germany) at a heating rate of 5 °C/min. The samples were sized at 5 mm×5 mm×25 mm, and their thermal expansion behavior was calculated using:

$$\alpha = \frac{1}{L} \frac{dL}{dt} \quad (1)$$

The initial length of the sample, denoted as L, was measured. Both the metal lead 4J50/Cu and the glass-ceramic samples were tested to determine their glass transition point and softening point before and after precipitation. The testing conditions were as follows: the temperature range for the metal lead was from room temperature to 900 °C, while for the glass, it ranged from room temperature to 700 °C.

The wettability of the sample on an Al₂O₃ ceramic substrate was evaluated using a CJY-II-1600 sintering imager. The temperature increase rate was set to 10 °C/min, allowing the imager to visually capture the wettability of the sample's components.

Microhardness was measured using an HV-1000 microhardness tester, applying a 1 kg load for a duration of 20s. Between 5 and 8 measurements were taken for each specimen, and the average value was calculated using the following formula:

$$H_v = \frac{1854.4P}{(2a)^2} \quad (2)$$

where H_v is the hardness value (GPa); P is the applied load (N); and 2a is the diagonal length (μm).

The flexural strength of the specimen was measured using the three-point bending method. Each specimen was machined to dimensions of 5 mm×5 mm×25 mm, with its surface polished and chamfered. The tests were conducted on a WSM-10KN universal testing machine at a loading rate of 0.5 mm/min. The results were obtained by averaging the flexural strength values from four samples. The flexural strength was calculated using the following formula:

$$\sigma = \frac{3PL}{2bh^2} \quad (3)$$

Where σ is the breaking strength (MPa); P is the load (N); L is the span (mm); b is the width of the specimen (mm); and h is the height of the specimen (mm).

Fig. 1 (b) illustrates the sealing assembly and the associated sealing temperature regime. Preparation of the glass insulator involved weighing 10g of LZAS glass powder and placing it in a mortar. Then, 5ml of alcohol was added, and the mixture was stirred until adhesive particles formed. An appropriate amount of the mixture was pressed into a hollow mold. After shaping, the sample was sintered in a muffle furnace at a temperature increase rate of 10°C/min, reaching 360°C and holding this temperature for 30min.

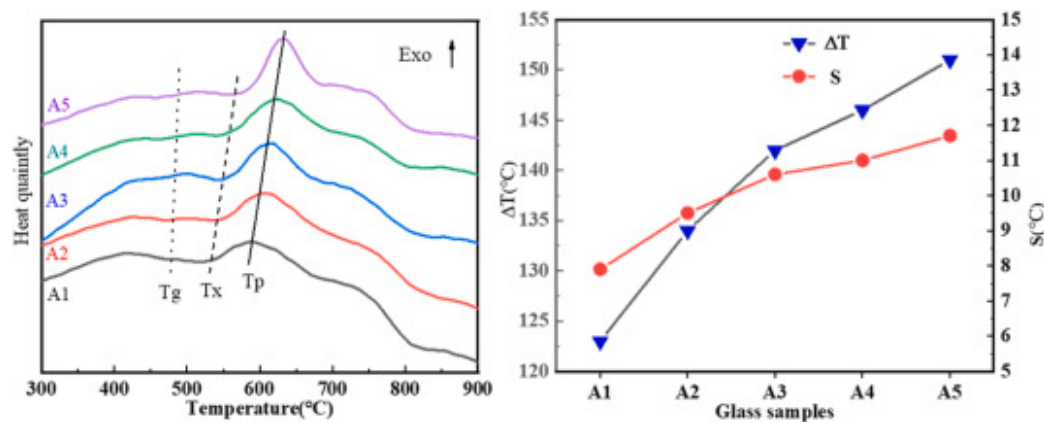
Alumina ceramic was used as the base, and a hole approximately 2.1 mm in diameter was drilled into it to secure the 4J50/Cu lead and the glass insulator. The components were sealed in a tube furnace connected to nitrogen. The temperature was increased at a rate of 10°C/min until 930°C, where the sample was held for 20min before cooling naturally.

Afterward, the sample was cut horizontally, polished until smooth, and fixed onto a test platform using conductive glue. The surface was sputtered with gold, and the sealed sample was characterized using a field emission scanning electron microscope (SEM Japan Electronics JSM-IT200). Additionally, the sealed interface was analyzed using an energy dispersive spectrometer (EDS Japan Electronics JSM-IT200) for elemental line scanning.

3. Results and discussion

3.1. Basic glass DSC analysis

Fig. 2 illustrates the DSC curves of the A# series glass. The samples all exhibit one endothermic valley and one well-defined exothermic peak, namely glass transition temperature (T_g) and crystallization temperatures (T_p). The T_g of the A# series glass ranges from 460°C to 480°C, during which the glass structure undergoes significant rearrangement, enabling nucleation. The crystallization peak temperature T_p of the A# series glass lies between 590°C and 630°C. In this range, small crystals start to form within the glass.



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Fig. 2. DSC and GFA curves of A# series glasses.

The crystallization peak temperature T_p increases as the $\text{Al}_2\text{O}_3/\text{ZnO}$ ratio rises. This trend may be attributed to the higher field strength of Zn^{2+} ions, which exerts an ordering effect on surrounding oxygen ions, potentially inducing phase separation [22]. In contrast, Al_2O_3 stabilizes the glass system [23].

The glass-forming ability (GFA) was determined using the Dietzel and Saad-Paulin methods [24,25], with results presented in the corresponding table. These equations are as follows:

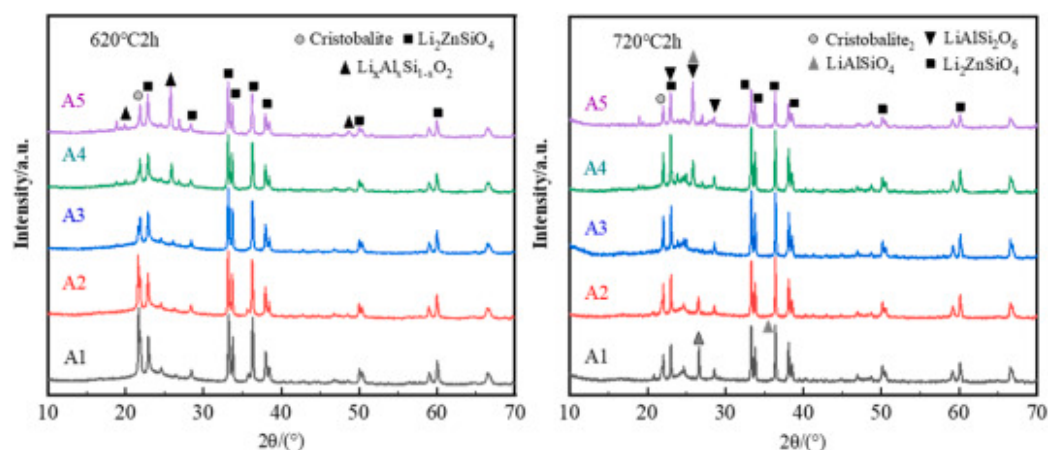
$$\text{Dietzel : } \Delta T = (T_p - T_g) \quad (4)$$

$$\text{Saad - Paulin : } S = (T_p - T_x)(T_x - T_g) / T_g \quad (5)$$

The parameters ΔT and S are used to characterize the GFA (glass-forming ability). Higher ΔT and S values indicate a greater tendency for glass formation. The GFA curve demonstrates that as the $\text{Al}_2\text{O}_3/\text{ZnO}$ ratio increases, the ΔT and S values rise significantly, thereby improving the stability of the glass network.

3.2. Crystallization behavior and microstructure of glass-ceramics

Fig. 3 illustrates the XRD patterns of the base glasses with varying $\text{Al}_2\text{O}_3/\text{ZnO}$ ratios after crystallization at 620°C and 720°C for 2h. All five samples exhibit the presence of $\text{Li}_2\text{ZnSiO}_4$ (JCPDF 4-001-8424) and SiO_2 (JCPDF 1-073-3456). As ZnO is gradually and fully replaced by Al_2O_3 , $\text{Li}_x\text{Al}_x\text{Si}_{1-x}\text{O}_2$ (JCPDF 040-0073) is precipitated in samples A4 and A5. Heat treatment at 720°C , however, leads to the formation of LiAlSiO_4 (JCPDF 04-008-2032) in samples A1 and A2. Additionally, LiAlSiO_4 (JCPDF 026-0839) and $\text{LiAlSi}_2\text{O}_6$ (JCPDF 035-0797) are observed in samples A4 and A5. The disappearance of the LiAlSiO_4 peaks in samples A1 to A3 may result from the formation of $\text{Li}_x\text{Al}_x\text{Si}_{1-x}\text{O}_2$, which occurs between LiAlSiO_4 and SiO_2 . Meanwhile, the presence of $\text{LiAlSi}_2\text{O}_6$ in samples A4 and A5 suggests that the formed $\text{Li}_x\text{Al}_x\text{Si}_{1-x}\text{O}_2$ has undergone further substitution. Additionally, as the $\text{Al}_2\text{O}_3/\text{ZnO}$ ratio increases, the diffraction peak intensities of SiO_2 and $\text{Li}_2\text{ZnSiO}_4$ in the samples tend to diminish. This indicates that higher Al^{3+} content promotes the formation of $\text{Li}_x\text{Al}_x\text{Si}_{1-x}\text{O}_2$ and $\text{LiAlSi}_2\text{O}_6$ phases.



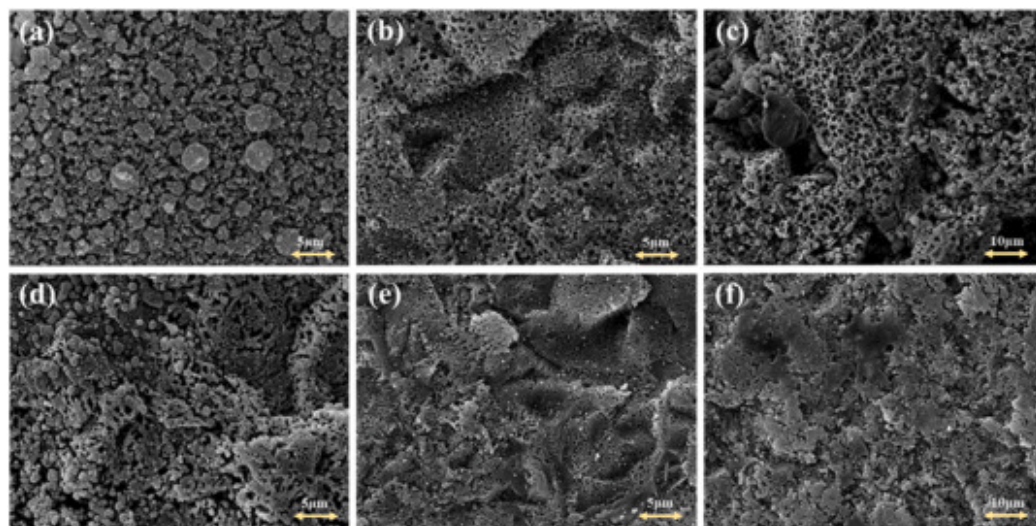
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Fig. 3. A# XRD plots of precipitated crystals at 620°C and 720°C for 2h

A detailed analysis of the crystallization mechanism reveals that the quartz phase, observed at $600\text{--}700^\circ\text{C}$, predominantly exhibits a hexagonal crystal system characteristic of the β -quartz phase. This phase features an open hexagonal helical structure composed of numerous interconnected silica-oxygen tetrahedra [26]. When Al^{3+} replaces Si^{4+} within the silica-oxygen tetrahedra, Li^+ or Zn^{2+} ions occupy the voids to balance the space charge [27]. When half of the Si^{4+} is replaced by Al^{3+} , β -eucryptite forms. Similar to β -quartz, β -eucryptite also has a hexagonal helical structure, and the two can form a continuous solid solution, resulting in the development of a β -quartz solid solution [28].

Fig. 4 presents scanning electron microscope (SEM) images of samples A1, A3, and A5 after crystallization at 620°C and 720°C for 2h. Figure (a) depicts numerous spherical crystals, while Figures (b) and (c) exhibit honeycomb-like structures. Based on the XRD patterns, at 620°C, the composition of the $\text{Al}_2\text{O}_3/\text{ZnO}$ component alters, leading to a transformation in the crystal phase from spherical $\text{Li}_2\text{ZnSiO}_4$ crystals to a morphology rich in $\text{Li}_x\text{Al}_x\text{Si}_{1-x}\text{O}_2$, with a prominent honeycomb structure. This indicates the formation of an increased amount of $\text{Li}_x\text{Al}_x\text{Si}_{1-x}\text{O}_2$.



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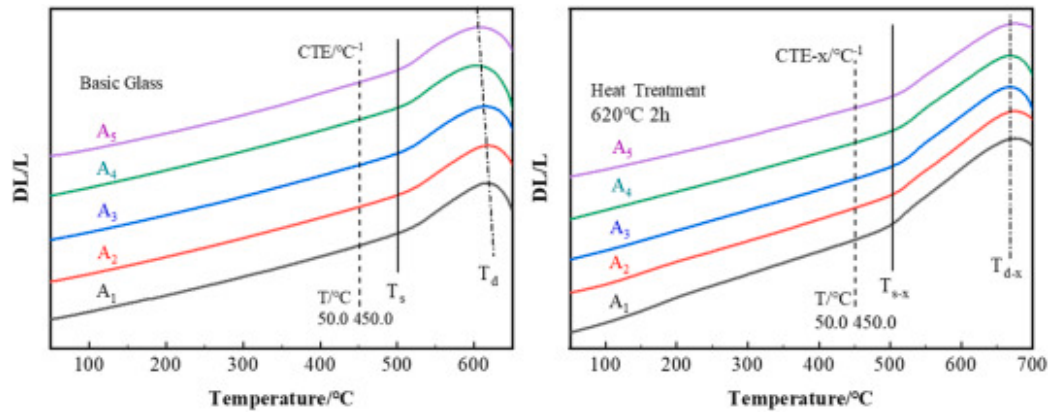
Fig. 4. Microstructure of glass-ceramic (a), (b), and (c) represent the SEM images of A1, A3 and A5 after crystallization at 620°C for 2h; (d), (e) and (f) represent the SEM images of A1, A3 and A5 after crystallization at 720°C for 2h.

At a sintering temperature of 720°C, the samples undergo densification, resulting in fine dendrites forming an interconnected network. The smaller grains represent newly precipitated crystals, while the larger ones are grown-up crystals. This observation suggests that nucleation and crystal growth occur simultaneously, without distinct boundaries between the two processes.

3.3. Properties of glass-ceramic

Fig. 5 illustrates the coefficient of thermal expansion for both the base glass and the glass heat-treated at 620°C for 2h. T_S represents the expansion transition point of the glass, T_d denotes the softening temperature point of the glass, CTE represents

the coefficient of thermal expansion of the glass from 50°C to 450°C, T_{s-x} is the expansion transition point of the heat-treated glass, T_{d-x} is the softening temperature point of the heat-treated glass, and CTE-x is the coefficient of thermal expansion of heat-treated glass from 50°C to 450°C. After heat treatment, there is minimal change in T_{s-x} , but a significant increase in T_{d-x} is observed.



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Fig. 5. Thermal expansion curves of the A# base glass and heat-treated glass.

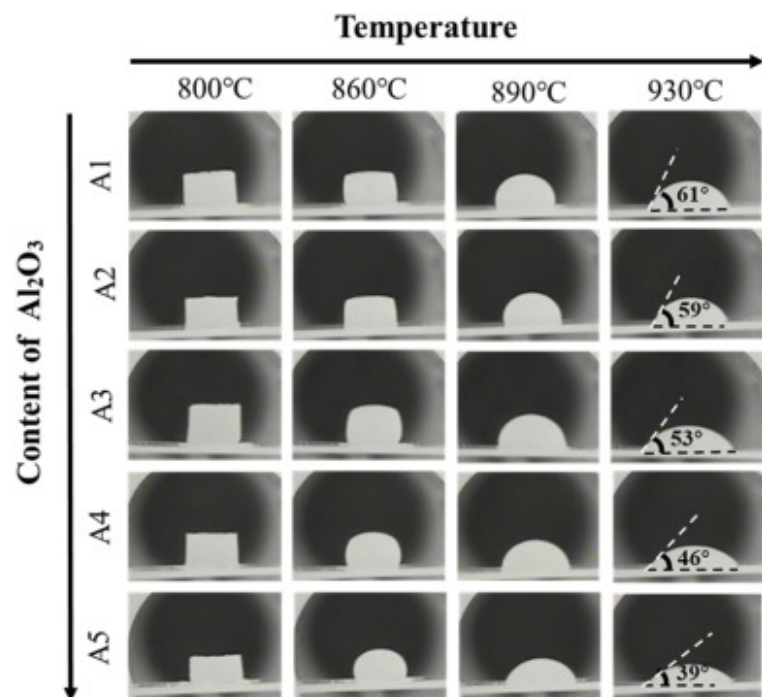
Table 3 reveals that variations in the Al₂O₃/ZnO ratio have negligible impact on the coefficient of thermal expansion of the base glass. However, the coefficient of thermal expansion of the glass-ceramic decreases as the Al₂O₃/ZnO ratio increases. This reduction is attributed to the crystalline phases present in the glass-ceramic. Specifically, the appearance of $\text{Li}_x\text{Al}_x\text{Si}_{1-x}\text{O}_2$, which shares structural similarities with LiAlSiO_4 and exhibits low or even negative expansion characteristics, accounts for the decrease in thermal expansion in the A# series samples after crystallization.

Table 3. Glass characteristics temperature and thermal expansion coefficient of the A# base glass and heat-treated glass.

Number	$T_s(^{\circ}\text{C})$	$T_{s-x}(^{\circ}\text{C})$	$T_d(^{\circ}\text{C})$	$T_{d-x}(^{\circ}\text{C})$	CTE($^{\circ}\text{C}^{-1}$)	CTE-x($^{\circ}\text{C}^{-1}$)
A1	519	500	617	676	10.9×10^{-6}	13.6×10^{-6}
A2	518	499	618	675	11.0×10^{-6}	12.4×10^{-6}

Number	Ts(°C)	Ts-x(°C)	Td(°C)	Td-x(°C)	CTE(°C-1)	CTE-x(°C-1)
A3	514	513	615	669	11.0×10^{-6}	11.8×10^{-6}
A4	514	514	604	669	11.2×10^{-6}	11.2×10^{-6}
A5	512	513	608	676	10.8×10^{-6}	10.2×10^{-6}

[Fig. 6](#) illustrates the temperature progression of the glass body as it rises from room temperature to 930°C at a heating rate of 5°C/min. The samples exhibit varying degrees of shrinkage at 800°C, and by 860°C, the edges and corners of the sintered bodies in samples A3 to A5 disappear, resulting in a rounded top. At this stage, the LZAS glass-ceramic undergoes crystallization followed by sintering, significantly increasing the temperature required for sample wetting. [Fig. 6](#) shows the wetting angles of the samples at 930°C, indicating improved wettability of the sintered body with higher alumina content. The increased Al₂O₃ content inhibits the crystallization ability of the LZAS glass-ceramic, enhances the glass phase content, and improves its high-temperature fluidity.

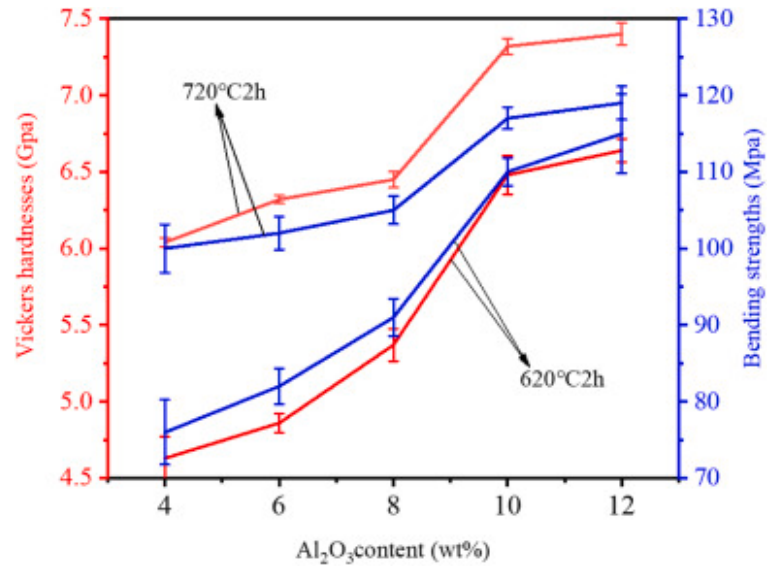


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Fig. 6. Comparison of high temperature thermal images of different specimens.

Fig. 7 illustrates the Vickers hardness and flexural strength of the glass-ceramic samples after crystallization at 620°C and 720°C for 2 h. The glass-ceramics crystallized at 720°C exhibit higher strength and hardness. As the Al₂O₃ content increases, the glass network becomes more rigid, which aligns with the previously discussed XRD results. The enhanced binding strength of Li_xAl_xSi_{1-x}O₂ contributes to the observed increase in hardness. Additionally, the rise in Al₂O₃ content improves the flexural strength of the glass-ceramic, which is attributed to higher crystallinity and density.



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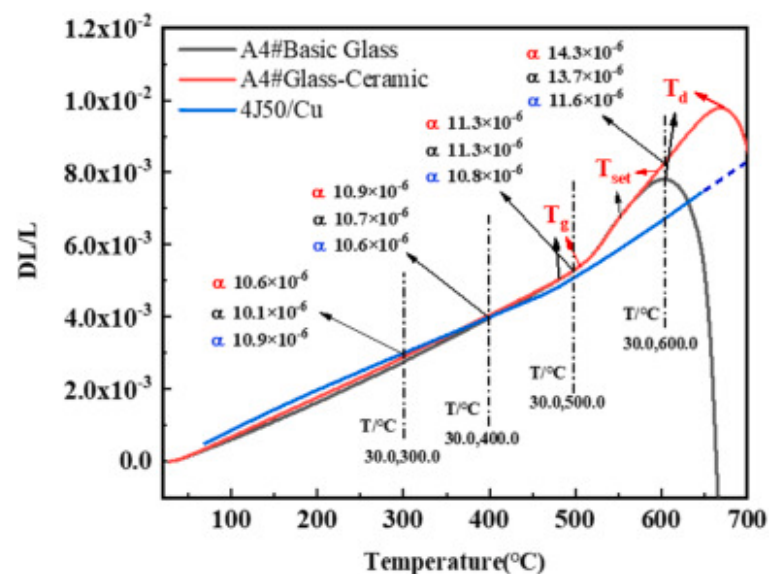
Fig. 7. Vickers hardness and flexural strength of A# glass-ceramic after crystallization at 620°C and 720°C for 2h.

3.4. LZAS glass-ceramic - 4J50/Cu lead seal performance

The coefficient of thermal expansion (CTE) plays a critical role in glass or glass-ceramic-metal seals, as stress arises from differential shrinkage during cooling after sealing. Fig. 8 displays the thermal expansion curves of 4J50/Cu and A4# glass, both before and after crystallization, allowing for the estimation of thermal strain caused by differential shrinkage. The thermal strain and the corresponding thermal stress due to differences in CTE can be calculated using the following formula.

$$\epsilon_M = \int_T^{T_0} (\alpha_2 - \alpha_1) dT \quad (6)$$

$$\sigma_M = \frac{E}{1-\mu} \epsilon_M \quad (7)$$



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Fig. 8. Coefficient of thermal expansion of A4# basic glass, A4# glass-ceramic and 4J50/Cu lead.

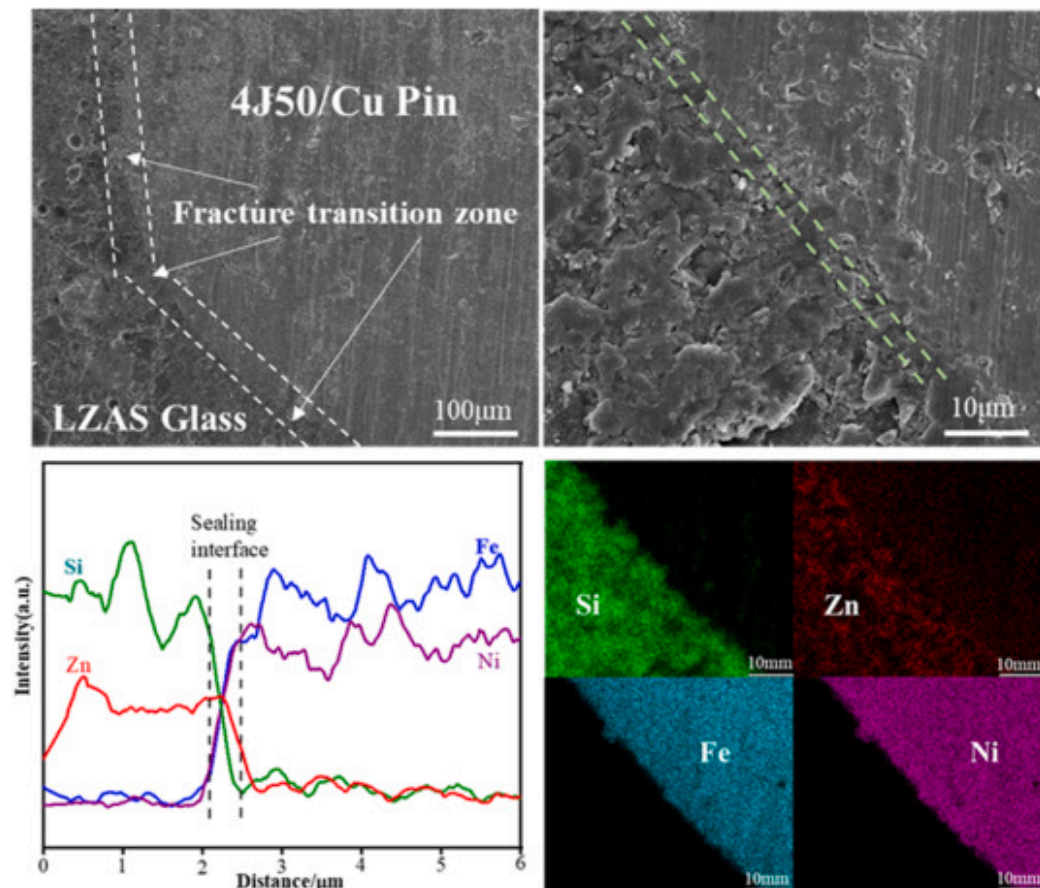
The temperature change from T_0 to T , along with the thermal expansion coefficients of the two materials (both of which are temperature-dependent functions), determines the thermal strain. Additionally, E represents the elastic modulus, and μ represents Poisson's ratio of the material. Based on Fig. 8 and the formula, the thermal expansion coefficients of A4# glass before and after crystallization align well with those of 4J50/Cu. As a result, the residual stress generated during cooling at high temperatures is minimal, meeting the requirements for matching and sealing with 4J50/Cu.

However, as the sealing temperature decreases—from a high sealing temperature to a lower temperature—a point is reached where the stress relaxation of the LZAS glass-ceramic becomes very slow. At this point, the elastic modulus and Poisson's ratio of the LZAS glass-ceramic are no longer constant, making it impossible to accommodate differences in thermal shrinkage between the 4J50/Cu lead and the LZAS glass-ceramic. A further temperature drop leads to the development of permanent stress over the relevant timescale [29].

As depicted in Fig. 8, the temperature range where stress relaxation can no longer fully accommodate shrinkage stress is referred to as the "set" point of the glass, T_{set} . This point is not a fixed temperature but occurs within a specific range. In the temperature interval between the softening point (T_d) and the transition point (T_g) of the glass, the LZAS microcrystalline glass exhibits a broader stage for stress relaxation. Failures during this stage may occur during cooling or, more critically, as a result of static fatigue later in the part's lifecycle, which can compromise the seal.

It is speculated that the precipitation of crystals in LZAS microcrystalline glass retards the flow of the residual glass phase. At the end of high-temperature sealing, the cryogenic phase is held in place for a sufficient duration to mitigate potential hazards that could arise.

As shown in Fig. 9, which presents the SEM and EDS spectra of the cross-section of the LZAS glass and 4J50/Cu seal at 100× and 2000× magnification, all samples exhibit a certain degree of elemental diffusion and migration. At the LZAS glass sealing interface, elements Si and Zn participate in diffusion, whereas elements Fe and Ni diffuse at the 4J50/Cu lead interface. The transition layer of Zn within the LZAS glass is particularly prominent, indicating the formation of oxides within the sealing joint layer.



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Fig. 9. SEM and EDS mapping of LZAS glass-ceramic and 4J50/Cu lead.

During the cooling stage, once sealing is complete, the bonded components are not subjected to external forces such as extrusion or stress failure caused by large differences in thermal expansion. It is speculated that the transition layer of the seal is reinforced by an intermediate compound formed to connect the LZAS glass to the surface or oxide layer of the 4J50/Cu lead [30].

4. Conclusion

The sealing potential of glass-ceramic materials in the Li₂O-ZnO-Al₂O₃-SiO₂ (LZAS) system for 4J50/Cu metal leads was investigated. Increasing the Al₂O₃/ZnO ratio enhances the thermophysical properties of the base glass, including T_g, T_p, T_s, and T_d, while improving the stability of the glass network. LZAS glass-ceramics containing crystalline phases such as lithium zinc silicate and cristobalite were synthesized using the sintering method at 620°C and 720°C.

At 620°C, samples A4 and A5 exhibit a negative expansion coefficient and a robust honeycomb-like β-quartz structure. At 720°C, β-eucryptite and β-spodumene crystal phases with negative expansion coefficients reappear, significantly enhancing the strength of the LZAS glass-ceramics. While the increase in the Al₂O₃/ZnO ratio has no effect on the coefficient of thermal expansion of the base glass, the coefficient of thermal expansion of the microcrystalline glass decreases substantially.

LZAS glass-ceramics demonstrate excellent wettability at 930°C, and sample A4 exhibits a coefficient of thermal expansion that matches that of the 4J50/Cu lead wires, creating a reliable sealing interface. However, the standard high-temperature sealing temperature regime is unsuitable for LZAS microcrystalline glass. Holding the glass for an adequate duration at a lower temperature phase may be necessary to mitigate potential risks and ensure sealing integrity.

CRedit authorship contribution statement

Hongwei Guo: Writing – original draft, Supervision, Resources. **Yibo Gao:** Writing – original draft, Investigation, Formal analysis, Data curation. **Shouyi Guo:** Investigation. **Mi Liu:** Methodology. **Dong Nie:** Investigation. **Zhilong Zhao:** Resources. **Yuxuan Gong:** Writing – original draft. **Relva C. Buchanan:** Methodology.

Declaration of interest statement

The authors state that they have no known competing financial interests or personal relationships at this time and will not affect the work reported in this article.

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